# Menstrual Cycle Prediction System using Machine Learning

## 1. Introduction

The Menstrual Cycle Prediction System is a machine learning–based solution designed to estimate key reproductive events such as ovulation date, next period date, and fertile window. The system combines **data-driven prediction (machine learning)** with **biological rules** to produce accurate and interpretable outputs.

This hybrid approach ensures both **predictive capability** and **biological validity**, which is critical in health-related applications.

## 2. Problem Statement

The objective of this project is to:

- Predict **ovulation day** using historical cycle data
- Estimate **next menstrual period date**
- Identify the **fertile window**

Challenges include:

- Variability in menstrual cycles across individuals
- Noisy and inconsistent real-world data
- Need for biologically meaningful predictions

## 3. System Architecture

The system follows a structured pipeline:

```
Raw Dataset
     ↓
Data Preprocessing
     ↓
Feature Selection
     ↓
Machine Learning Model
     ↓
Ovulation Prediction
     ↓
Biological Rule Engine
     ↓
Final Predictions
```

### Outputs:

- Ovulation Date
- Next Period Date
- Fertile Window

## 4. Technologies Used (Tech Stack)

### 4.1 Programming Language

- **Python**
    - Used for implementing the entire pipeline due to its strong ecosystem for data science and machine learning.

### 4.2 Data Processing Layer

- **Pandas**
- **NumPy**

**Purpose:**

- Load and manipulate dataset
- Handle missing values

- Perform feature engineering

## 4.3 Machine Learning Layer

- **Scikit-learn**

**Model Used:**

- **Random Forest Regressor**

**Reason for Selection:**

- Handles nonlinear relationships effectively
- Performs well on small tabular datasets
- Reduces overfitting using ensemble learning

## 4.4 Feature Scaling

- **StandardScaler (Scikit-learn)**

**Purpose:**

- Normalize feature values
- Improve model stability

## 4.5 Model Persistence

- **Joblib**

**Purpose:**

- Save trained model (`.pkl` file)
- Save scaler for consistent predictions

### 4.6 Rule-Based Engine

- Implemented using **Python logic**

**Purpose:**

- Convert ovulation predictions into:
  - Next period date
  - Fertile window

### 4.7 Execution Environment

- Command Line Interface (CLI)
- Developed using **VS Code**

## 5. Dataset Description

The dataset consists of **1665 menstrual cycle records** with 80 features.

### Key features used:

- CycleNumber
- LengthofCycle
- LengthofLutealPhase
- TotalNumberofHighDays
- TotalNumberofPeakDays
- Age
- BMI
- EstimatedDayofOvulation (target variable)

## 6. Data Preprocessing

### Steps performed:

1. **Data Loading**

    a. CSV dataset loaded using Pandas
2. **Data Cleaning**
    a. Converted string-based numeric columns to numeric format
    b. Removed invalid and missing values
3. **Feature Selection**
    a. Reduced 80 features to 7 relevant biological features
    b. Eliminated noise and irrelevant attributes
4. **Final Dataset Creation**
    a. Clean dataset saved as `cleaned_cycle_data.csv`

# 7. Model Training

### Steps:

1. **Train-Test Split**
    a. 80% training data
    b. 20% testing data
2. **Feature Scaling**
    a. Applied StandardScaler
3. **Model Training**
    a. Random Forest Regressor trained on processed data
4. **Evaluation Metrics**
    a. Mean Absolute Error (MAE)
    b. Mean Squared Error (MSE)

### Performance:

- MAE ≈ 1–2 days
- Indicates high prediction accuracy for biological data

# 8. Prediction System

### Step 1 — ML Prediction

The trained model predicts:

```
Cycle Features → Ovulation Day
```

### Step 2 — Rule-Based Calculations

Using biological principles:

***Next Period Date:***

```
Next Period = Ovulation Date + Luteal Phase Length
```

***Fertile Window:***

```
Fertile Window = Ovulation – 5 days → Ovulation + 1 day
```

## 9. Output Example

Input:

- Last period date: 2026-03-01
- Cycle length: 29
- Luteal phase: 14

Output:

- Ovulation Date: March 16
- Next Period Date: March 30
- Fertile Window: March 11 – March 17

## 10. Project Structure

```
menstrual-cycle-predictor/

dataset/
    menstrual_cycle.csv
    cleaned_cycle_data.csv
```

```
model/
    cycle_prediction_model.pkl
    scaler.pkl

scripts/
    preprocess_dataset.py
    train_cycle_model.py
    predict_cycle.py
```

## 11. Key Insights

- Ovulation is the **central variable** in cycle prediction
- Period prediction is **not purely ML-based**, but partially deterministic
- Hybrid systems (ML + rules) are more reliable for biological problems

## 12. Limitations

- Treats each cycle independently (no time-series modeling)
- Limited dataset size
- Does not handle irregular cycles explicitly

## 13. Future Improvements

- Time-series models (LSTM, Transformers)
- Personalized user modeling
- Confidence scoring for predictions
- Web/mobile application interface
- Integration with real-time user inputs

## 14. Conclusion

This project demonstrates a practical application of machine learning in healthcare by combining:

- Data-driven prediction
- Domain-specific biological knowledge

The system achieves **accurate and interpretable predictions**, making it a strong foundation for real-world menstrual health applications.